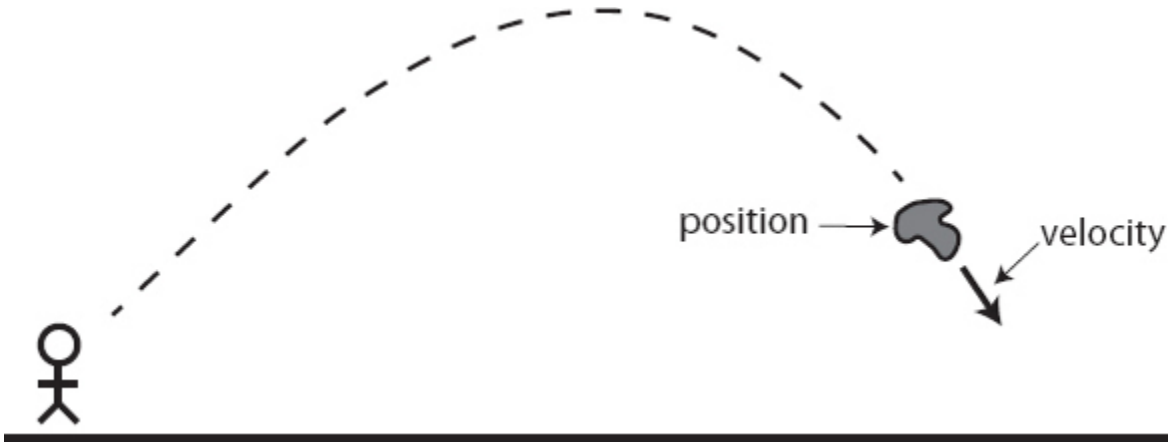
abstract form: the rock is an object, and that object has a *location in space*, and that location *changes with time*.

Classical mechanics tells us precisely how the position of the rock changes with time. We're very used to that by now, so it's worth reflecting on how impressive this is. Newton doesn't hand us some vague platitudes about the general tendency of rocks to move more or less in this or that fashion. He gives us exact, unbreakable rules for how everything in the universe moves in response to everything else—rules that can be used to catch baseballs or land rovers on Mars.

Here's how it works. At any one moment, the rock will have a position and also a velocity, a rate at which it's moving. According to Newton, if no forces act on the rock, it will continue to move in a straight line at constant velocity, for all time. (Already this is a major departure from Aristotle, who would have told you that objects need to be constantly pushed if they are to be kept in motion.) If a force does act on the rock, it will cause acceleration —some change in the velocity of the rock, which might make it go faster, or slower, or merely alter its direction—in direct proportion to how much force is applied.



That's basically it. To figure out the entire trajectory of the rock, you need to tell me its position, its velocity, and what forces are acting on it. Newton's equations tell you the rest. Forces might include the force of gravity, or the force of your hand if you pick up the rock and throw it, or the force from the ground when the rock comes to land. The idea works just as well for billiard balls or rocket ships or planets. The project of physics, within this classical paradigm, consists essentially of figuring out what